JOHN CHANAS

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EDUCATION

The University of Waikato *November 2028 (Expected)*
B.E. (Honours) in Mechatronics Engineering

PROJECTS

Self-Balancing Robot July 2026 - Present
Arduino, 3D Printing, Embedded C++ *Hamilton, Waikato*

- Designed and 3D-printed a custom robot chassis in Fusion 360 to integrate the mechanical and electronic components.
- Developed a dual-voltage power system integrating an IMU, motor driver and microcontroller through I2C and PWM connections.

Op-Amp Circuit Design March 2026 - July 2026
Differentiators, Inverters, Thermostat and Radio Mic *University of Waikato*

- Built an LM358 temperature-triggered comparator, calculating NTC thermistor values to determine switching behaviour.
- Built and soldered an FM transmitter circuit using an op-amp, calculating the oscillation frequency and antenna power.
- Designed and simulated inverting, non-inverting and differentiator op-amp circuits using LTSpice.

Pomodoro Productivity Timer March 2026 - April 2026
Arduino, Embedded C++ *Hamilton, Waikato*

- Built a hardware productivity timer integrating an Arduino, OLED display, touch sensor, joystick and LED indicators.
- Programmed the Arduino in C++ to process user inputs and provide real-time visual feedback through multiple outputs.

Truss Support Frame August 2025 - October 2025
Structural Analysis, Load Testing *University of Waikato*

- Applied mechanical analysis methods to calculate support forces and design a Pratt truss for a 5 kg load requirement.
- Won the class competition by exceeding the 5 kg target and successfully holding the full load for under 12 seconds.

WORK EXPERIENCE / EXTRACURRICULARS

Millennium Hotel and Resort - Manuel's Taupō July 2022 - March 2023
Food and Beverage Assistant *Taupō, Waikato*

- Provided accurate and efficient food and beverage service to hotel guests in a busy front-of-house environment.
- Handled customer orders and service during high-volume periods while maintaining accuracy and service standards.
- Coordinated with kitchen and front-of-house staff to maintain efficient service during peak periods.

Burger King, Taupō August 2024 - March 2025
Crew Cook *Taupō, Waikato*

- Prepared food to required recipe, hygiene and quality standards during high-volume service periods.
- Managed multiple kitchen stations while maintaining speed, accuracy and food-safety standards.
- Worked collaboratively with crew members to keep orders moving efficiently during busy periods.

TECHNICAL STRENGTHS

Programming	Embedded C++, Python, MATLAB, C#
Engineering Software	Fusion 360, LTSpice, SAP2000
Electronics and Hardware	Arduino, Op-Amp Circuits, Oscilloscope, Soldering, 3D Printing
Other	Microsoft Office